

1. ① 由 d'Alembert 公式

$$u = \frac{1}{2}((x-3t)^3 + (x+3t)^3) + \frac{1}{6} \int_{x-3t}^{x+3t} \sin x dx$$

$$= x^3 + 27t^2x + \frac{1}{6}(\cos(x-3t) - \cos(x+3t))$$

② $u_{tt} - 9u_{xx} = x + xt$

$$u = \frac{xt^3}{6} - \frac{x^3}{54}$$

$$\begin{cases} u_{tt} = 9u_{xx} \end{cases}$$

$$\begin{cases} u|_{t=0} = \frac{55}{54}x^3 & u_x|_{t=0} = \sin x \end{cases}$$

$$u = \frac{xt^3}{6} - \frac{x^3}{54} + \frac{55}{54}(x^3 + 27t^2x) + \frac{1}{6}(\cos(x-3t) - \cos(x+3t))$$

② 或

$$\begin{cases} w_{xx} = 9w_{xx} \\ w = 0 \end{cases}$$

$$w_{t=\tau} = x + x\tau$$

$$\Rightarrow w(x, t, \tau)$$

$$= \frac{1}{6} \int_{x-3(t-\tau)}^{x+3(t-\tau)} (x + x\tau) dx$$

$$= \frac{1}{12}(\tau+1)((x+3t-3\tau)^2 - (x-3t+3\tau)^2)$$

$$= \frac{1}{12}(\tau+1)6(t-\tau) \cdot 2x$$

$$= (\tau+1)(t-\tau)x$$

故而特解为 $\int_0^t (\tau+1)(t-\tau)x d\tau$

$$= \int_0^t (-\tau^2 + (t-1)\tau + t)x d\tau$$

$$= -\frac{t^3}{3}x + \frac{t^2(t-1)}{2}x + t^2x$$

$$= \frac{t^3x}{6} + \frac{t^2}{2}x \quad \text{余下同上}$$

2. $u = a_2x^2 + a_1x + a_0$

$$u_{tt} - a^2 u_{xx} = 1$$

$$\Rightarrow a_2 = \frac{1}{2a^2}$$

$$u(0) = u_x(l) = 0$$

则 $a_0 = 0$ $(\frac{1}{2a^2}x^2 + bx)'_{x=l} = 0$ $\frac{x}{a^2} + b = 0$
 $\frac{l}{a^2} + b = 0$ $b = -\frac{l}{a^2}$

$\Rightarrow$ 特解 $u_1 = \frac{x^2}{2a^2} - \frac{l}{a^2}x$

$u = u_1 + u_2$

u_2 $\begin{cases} u_{tt} = a^2 u_{xx} \\ u_{x=0} = u_{x=l} = 0 \\ u(0, x) = \frac{l}{a^2}x - \frac{x^2}{2a^2} + \varphi(x) \\ u_t(0, x) = \psi(x) \end{cases}$

$a \sin vx + b \cos vx$
 $a(\sin vx)'$
 $\cos vx$

$\Rightarrow u = \sum_{n=0}^{+\infty} (A_n \cos(a(\frac{n+\frac{1}{2}}{l})\pi t) + B_n \sin(a(\frac{n+\frac{1}{2}}{l})\pi t)) (\cos \frac{(n+\frac{1}{2})\pi x}{l})$
 其中 $A_n = \frac{2}{l} \int_0^l (\frac{l}{a^2}x - \frac{x^2}{2a^2} + \varphi(x)) \cos \frac{(n+\frac{1}{2})\pi x}{l} dx$
 $a \frac{(n+\frac{1}{2})\pi}{l} \cdot B_n = \frac{2}{l} \int_0^l \psi(x) \cos \frac{(n+\frac{1}{2})\pi x}{l} dx$

三.

$$\begin{cases} \Delta u = 0 \\ u|_{r=1} = 0, u|_{\theta=0} = 0 \\ u|_{\theta=\frac{\pi}{3}} = \varphi(r) \end{cases}$$

$$u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta} = 0$$

$$r^2 \frac{R''}{R} + r \frac{R'}{R} = -\frac{\Theta''}{\Theta} = -\lambda$$

$$\Rightarrow R'' \text{ 的解 } w = \ln r$$

$$\text{则得 } R''(w) + \lambda R(w) = 0$$

$$\frac{\Theta''}{\Theta} - \lambda = 0$$

$$R(0) = R(1) = 0$$

$$\Rightarrow \lambda \text{ 为 } (n\pi)^2 \text{ 解 } \sin(n\pi \ln r)$$

$$\text{故而解为 } \sum_{n=1}^{+\infty} (A_n \cosh(n\pi \theta) + B_n \sinh(n\pi \theta)) \sin(n\pi \ln r)$$

$$\Rightarrow \text{原式} = \sum_{n=1}^{+\infty} A_n \sinh(n\pi \theta) \sin(n\pi \ln r)$$

$$\text{故而 } A_n = \frac{\sinh \frac{n\pi^2}{3}}{2 \int_1^{\frac{1}{3}} \frac{1}{r} \sin(n\pi \ln r) \varphi(r) dr}$$

$$\Rightarrow \text{原式 } A_n = \frac{2 \int_1^{\frac{1}{3}} \frac{1}{r} \sin(n\pi \ln r) \varphi(r) dr}{\sinh \left(\frac{n\pi^2}{3} \right)}$$

$$\rho(x) = e^{-\int \frac{b_1(x)}{b_2(x)} dx} = \frac{1}{r}$$

112.

$$\begin{cases} \Delta_3 u = 0 \\ u_r = \frac{1}{r} u_{rr} = 0 \end{cases}$$

$$u|_{z=0} = r - r^2$$

$$u_z = 0$$

关于 φ 无关系
轴对称

$$u_{rr} + \frac{1}{r} u_r + u_{zz} = 0$$

$$\Rightarrow \frac{z''}{z} = \frac{R''}{R} + \frac{1}{r} \frac{R'}{R} = -\lambda$$

$$R = A J_0(\omega r) + B N_0(\omega r)$$

$$1 \quad \text{因为 } r=0 \text{ 有奇点}$$

$$\Rightarrow R = J_0(\omega r) A$$

$$\lambda = \omega_n^2$$

$$J_0(\omega_n) = 1$$

$$\Rightarrow \text{解}$$

$$\sum_{n=1}^{+\infty} (A_n \exp^{-\omega_n z} + B_n \exp^{\omega_n z}) J_0(\omega_n r)$$

$$A_n \exp^{-2\omega n} + B_n \exp^{2\omega n} = 0$$

$$A_n + B_n$$

$$= \frac{\int_0^1 r J_0(\omega r) (r-r^2) dr}{\frac{1}{2} J_1^2(\omega)}$$

$$= \frac{\int_0^\omega x J_0(x) (\omega x - x^2) dx}{\frac{\omega^4}{2} J_1^2(\omega)}$$

$$= \frac{\int_0^\omega (\omega x - x^2) dx J_1(x)}{\frac{\omega^4}{2} J_1^2(\omega)}$$

$$= \frac{-\int_0^\omega (2x - \omega) x J_1(x) dx}{\frac{\omega^4}{2} J_1^2(\omega)}$$

五. ① $\frac{2}{3}P_2(x) + P_1(x) + \frac{4}{3}P_0(x)$

② $\int_{-1}^1 P_{2019}'(x) P_{2021}(x) dx$

$$= \int_{-1}^1 P_{2019}'(x) dP_{2021}(x)$$

$$= P_{2021}(1) P_{2019}'(1) - P_{2021}(-1) P_{2019}'(-1)$$

$$- \int_{-1}^1 P_{2019}''(x) P_{2021}(x) dx$$

$$= P_{2021}(1) P_{2019}'(1) - P_{2021}(-1) P_{2019}'(-1)$$

$$= 2 P_{2021}(1) P_{2019}'(1)$$

$$= 2 \frac{1}{2^{2021} 2021!} \left((2+\lambda)^{2021} e^{2021\lambda} \right)_{\lambda=0}^{(2021)} \times \left((2+\lambda)^{2019} e^{2019\lambda} \right)_{\lambda=0}^{(2019)} \frac{1}{2^{2019} 2019!}$$

$$= 2 \frac{1}{2^{2021} 2021!} 2^{2021} (2021)! \times 2020! 2^{2018} C_{2019}' \frac{1}{2^{2019} 2019!}$$

$$= 2020 \cdot 2019$$

六. ① $\begin{cases} u_t = u_{xx} + 5u_x \\ u(0, x) = f(x) \end{cases}$

$$\begin{cases} \bar{u} = (-\lambda^2 - 5i\lambda) \bar{u} \\ \bar{u}|_{t=0} = 1 \end{cases}$$

$$\Rightarrow \bar{u} = e^{-(\lambda^2 + 5i\lambda)t}$$

故得 $u = \frac{1}{2\sqrt{\pi t}} \exp\left(-\frac{(x+5t)^2}{4}\right)$

$$② \int_0^t \frac{1}{2\sqrt{\pi\tau}} \int_{-\infty}^{+\infty} \exp\left(-\frac{(s+5\tau)^2}{4}\right) f(t-\tau-\lambda-s) ds d\tau$$

$$+ \int_{-\infty}^{+\infty} \varphi(s) \frac{1}{2\sqrt{\pi t}} \exp\left(-\frac{(x-s)^2}{4t}\right) ds$$

七. ① $u_0(x_0, y_0, z_0)$ $u_1(2-x_0, y_0, z_0)$

故而解为 $\frac{1}{4\pi} \left(\frac{1}{\sqrt{(x-x_0)^2 + (y-y_0)^2 + (z-z_0)^2}} - \frac{1}{\sqrt{(x+x_0-2)^2 + (y-y_0)^2 + (z-z_0)^2}} \right)$

② $z' = \frac{z}{3}$

$$u_{xx} + u_{yy} + u_{z'z'} = 0$$

$$u|_{x=0} = \varphi(y, 3z')$$

而 Green 函数解为

$$\frac{1}{4\pi} \left(\frac{1}{\sqrt{(x-x_0)^2 + (y-y_0)^2 + (z'-z_0)^2}} - \frac{1}{\sqrt{(x+x_0)^2 + (y-y_0)^2 + (z'-z_0)^2}} \right)$$

$$\text{故而 } \frac{\partial G}{\partial \vec{R}} = \left(\frac{\partial G}{\partial x_0}, \frac{\partial G}{\partial y_0}, \frac{\partial G}{\partial z_0} \right) (-1, 0, 0).$$

$$\text{故而原式} = \int \varphi(y_0, 3z_0') \frac{\partial G}{\partial x_0} dz_0' dy_0$$

$$+ \frac{2(x_0 - x) - 2(x_0 + x)}{4\pi (\sqrt{x^2 + (y - y_0)^2 + (z' - z_0')^2})^3}$$

$$= \frac{1}{2\pi} \int \varphi(y_0, 3z_0') \frac{x}{(\sqrt{x^2 + (y - y_0)^2 + (\frac{z}{3} - z_0')^2})^3} dz_0' dy_0$$